

Dense path
INVALID

Full score tensor
[B, H, S, S]

Global softmax
+ probabilities

Second $S \times S$
materialization

Does not fit
16 GB-class GPU

Streamed path
VALID

B=2 microbatch
16 ordered chunks

64×64 Q/KV tiles
online max + sum

Emit [B, S, D]
discard score tiles

6.56 GiB peak
no global $S \times S$